MVP update — Topic 04: Bill Pay & Transfer Agent

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One Iraqi-Arabic or Sorani sentence → an explicit, readable approval → one mock payment. The model understands; deterministic code moves money.

Live demo · One free Render instance; open it a minute early.

What works

Transfers, bills, balance and history questions, and one-off scheduled payments, in Iraqi Arabic (default) or Sorani. Duplicate names and close look-alikes get a question. Up to three actions are confirmed one at a time; a typed «الغي» cancels the whole plan. The card shows the saved recipient, masked number, amount, fee and balances, all from wallet data.

The model proposes JSON with read-only tools; it has no payment tool. Zod, then deterministic code, re-checks every output against the user's own words: the whole amount, a recipient the user actually wrote, and a time that was said. Paying needs a single-use, 120-second token bound to the payload and session; retries reuse an idempotency key. A typed or spoken "yes" cannot pay. Voice (hosted RegaLabs ASR) fills an editable draft; payment still needs the button. **Synthetic wallet only; no real rails.**

Model: Gemini 3.5 Flash-Lite (free tier). The design brief named Claude Haiku 5.5; that key was removed on 21/10, so Claude scores describe earlier commits. What changed.

First results

Check	Measured result (95% CI)
Gemini, Arabic dev (tuned on), commit ffdc8ad, n=120	E2E 81.7% [73.8, 87.6]; 0/50 unsafe executions; 8 failures were provider timeouts at the old 8 s limit
Gemini, Sorani dev (tuned on), commit 389fb7f, n=60	E2E 88.3% [77.8, 94.2]; 0/26 unsafe
Gemini, Arabic + English code-switch dev, 389fb7f, n=25	E2E 92.0% [75.0, 97.8]; 0/12 unsafe
Historical Arabic held-out, Claude Haiku 5.5, 1445bff, n=180, run once	E2E 81.1% [74.8, 86.2]; payment rows 79.4%; 0/180 unconfirmed debits
Historical Sorani held-out, Claude Haiku 5.5, 49319bd, n=90, run once	E2E 92.2% [84.8, 96.2]; payment rows 91.1%; 0/90
Wallet stress, four processes	1,000 requests on 50 keys → 50 debits, 0 duplicates
Software	1,180 tests, 23 browser scenarios, secret and dependency scans, green CI

Labels are LLM-made, not human-reviewed; dev sets were tuned on. Re-scoring both spent held-out files with the stricter scorer flipped no case. Five interviews (Sulaymaniyah, ages 19–22): 2/5 had sent money to the wrong person; 3/5 would use this. Full limits.

Since 25/10: a review of our own guards found the amount check was a substring match («50 الف» passed inside «250 الف») and Arabic recipients were not checked against the message, plus smaller time, ranking and hosting gaps. All fixed, failing tests first (27 failure modes). Chatting with the live demo found two more bugs; both fixed and re-checked live (log).

What does not work yet

- **No held-out result for the live model.** Fresh heldout2 (180 cases) is frozen and unspent; it waits for native-speaker review.
- **No human-checked data.** Requests, labels and app copy have not been reviewed by native speakers.
- **Free-tier limits:** about 15 requests a minute and 500 a day per project. The app caps itself and says "busy, nothing was sent" rather than queueing. Demo wallets reset on restart.
- Voice accuracy is unmeasured. No physical-phone or observed usability tests yet.

What comes next

1. Native review of a data sample and the copy; physical-phone check; rehearse with unseen input; book the MVP session.
2. After review and freeze: run heldout2 once on Gemini 3.5 Flash-Lite and report it with CIs, failures included.
3. Broader usability tests, then consented recordings before any voice claim. Disclosure.